

Departement Elektriese, Elektroniese en
Rekenaar-Ingenieurswese

Department of Electrical, Electronic and
Computer Engineering

Semestertoets 1

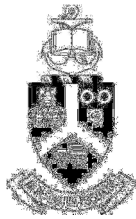
Kopiereg voorbehou

Module EBN122
Elektrisiteit en Elektronika
27 Augustus 2015

Semester Test 1

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Module EBN122
Electricity and Electronics
27 August 2015



UNIVERSITEIT VAN PRETORIA
UNIVERSITY OF PRETORIA
YUNIBESITHI YA PRETORIA

Eksamensinligting:

Exam information:

Maksimum punte: <i>Maximum marks:</i>	45	Volpunte: <i>Full marks:</i>	45
Duur van vraestel: <i>Duration of paper:</i>	90 minute <i>90 minutes</i>	Oopboek / toeboek: <i>Open / closed book:</i>	Toe <i>Closed</i>
Totale aantal bladsye (hierdie blad ingesluit): <i>Total number of pages (including this page):</i>		14	

BELANGRIK- IMPORTANT

- Die eksamenregulasies van die Universiteit van Pretoria geld.
The examination regulations of the University of Pretoria apply.
- Alle vrae moet op die ANTWOORDBLAD beantwoord word.
All questions must be answered on the ANSWER SHEET.
- Studente mag hul eie rowwe berekeninge op die vraestel doen – die vraestel word nie ingeneem nie.
Students may do their own rough calculations on the question paper – the question paper will not be taken in.
- Antwoorde moet eenhede insluit!
Answers must include units!
- Finale antwoorde moet as reële getalle geskryf word, en nie as breuke nie.**
Final answers must be written as real numbers, and not as fractions.

Dosent(e): <i>Lecturer(s):</i>	1 J Joubert
	2 A Chatterjee

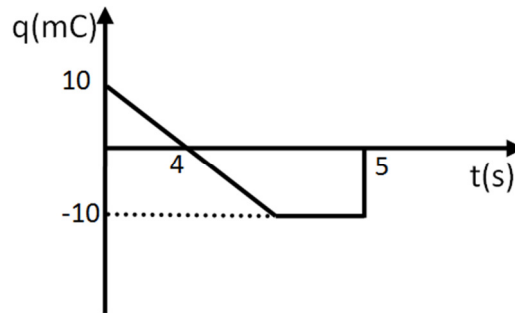
VRAAG/QUESTION 1 [22]

Vraag 1.1 [2]

Die lading deur 'n element word hieronder getoon.
Bepaal die stroom by $t = 4\text{s}$ (in mA).

Question 1.1 [2]

The charge through an element is shown below. Find
the current at $t = 4\text{s}$ (in mA).



Vraag 1.2 [2]

As 0.624×10^{18} elektrone deur 'n draad vloei in 5s, bepaal
die stroom (in mA).

Question 1.2 [2]

If 0.624×10^{18} electrons pass through a wire in 5s, find
the current (in mA).

Vraag 1.3 [2]

'n 15Ω weerstand is gekoppel aan 'n 10V battery. Hoeveel energie (in Joule) sal in die weerstand verkwis word in 1 minuut?

Question 1.3 [2]

A 15Ω resistor is connected to a 10V battery. How much energy (in Joule) will be dissipated in the resistor in 1 minute?

Vraag 1.4 [2]

Vir hoeveel ure kan jy 'n 250W TV gebruik vir 'n totale koste van 75 sent, as die koste van elektrisiteit R1/kWh is?

Question 1.4 [2]

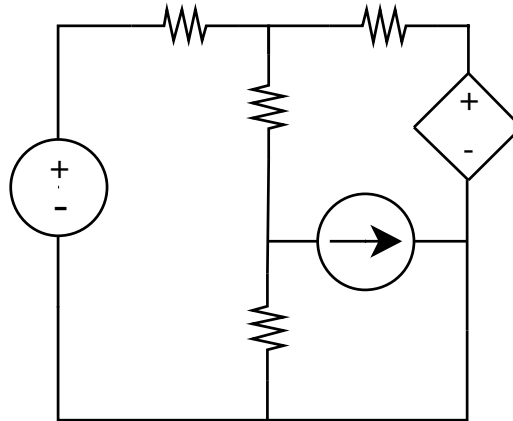
At electricity cost of R1/kWh, how many hours can you watch a 250W TV for a total cost of 75 cents?

Vraag 1.5 [2]

Bereken die aantal nodes en takke in die stroombaan.

Question 1.5 [2]

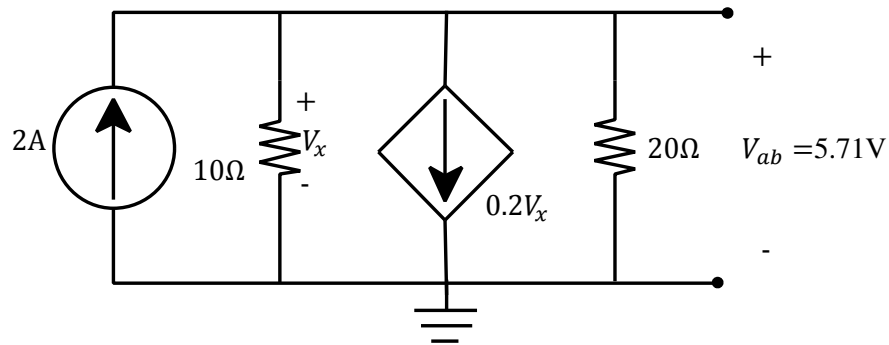
Determine the number of nodes and branches in the circuit.

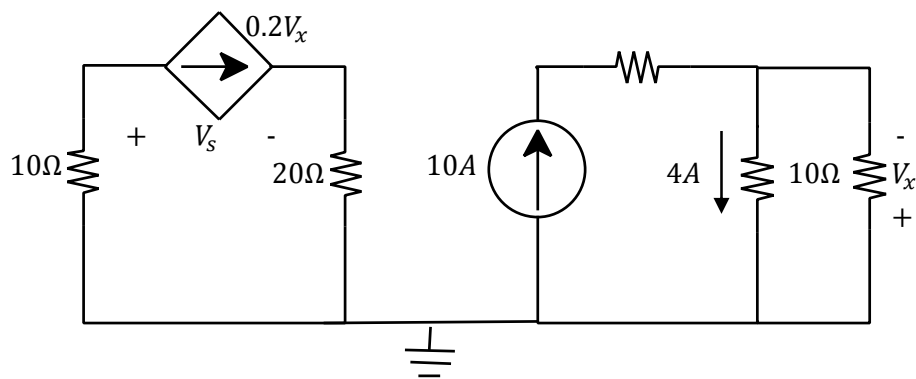
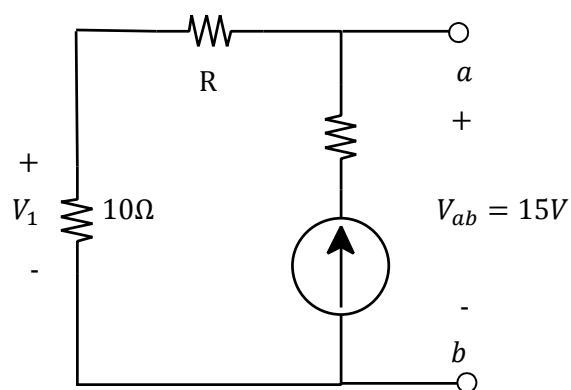
**Vraag 1.6 [2]**

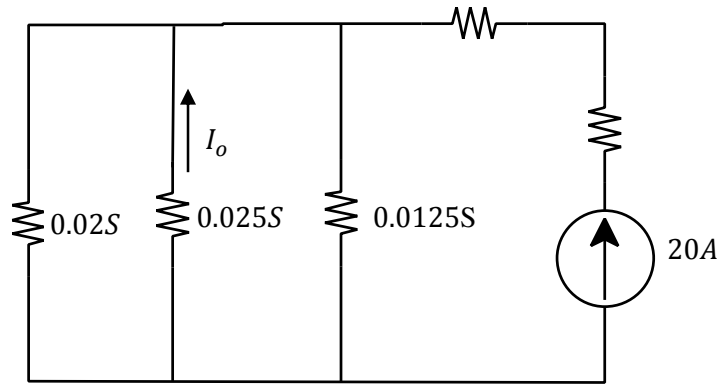
Bepaal die drywing ge-assosieer met die afhanklike stroombron.

Question 1.6 [2]

Find the power associated with the dependent current source.



Vraag 1.7 [2]Bepaal die spanning oor die afhanklike stroombron, V_s .**Question 1.7 [2]**Find the voltage across the dependent current source, V_s .**Vraag 1.8 [2]**Bepaal die waarde van R , sodat $V_I = 5V$.**Question 1.8 [2]**Find the value of R , such that $V_I = 5V$.

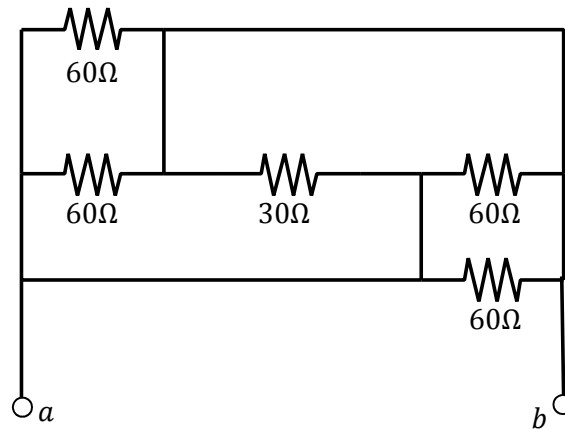
Vraag 1.9 [2]Bepaal die stroom I_o .**Question 1.9 [2]**Find the current I_o .

Vraag 1.10a [2]

Bepaal die ekwivalente weerstand tov terminale a-b.

Question 1.10a [2]

Find the equivalent resistance wrt terminals a-b.

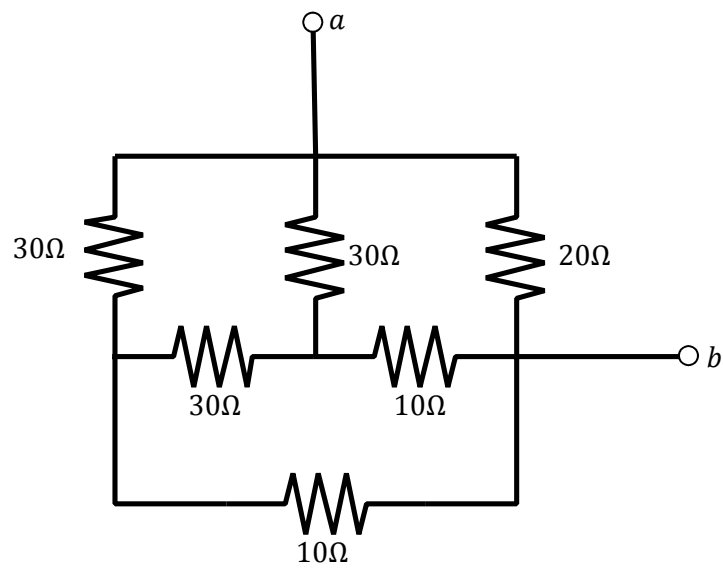


Vraag 1.10b [2]

Bepaal die ekwivalente weerstand tov terminale a-b.

Question 1.10b [2]

Find the equivalent resistance wrt terminals a-b.



VRAAG/QUESTION 2 [23]

Vraag 2.1 [4]

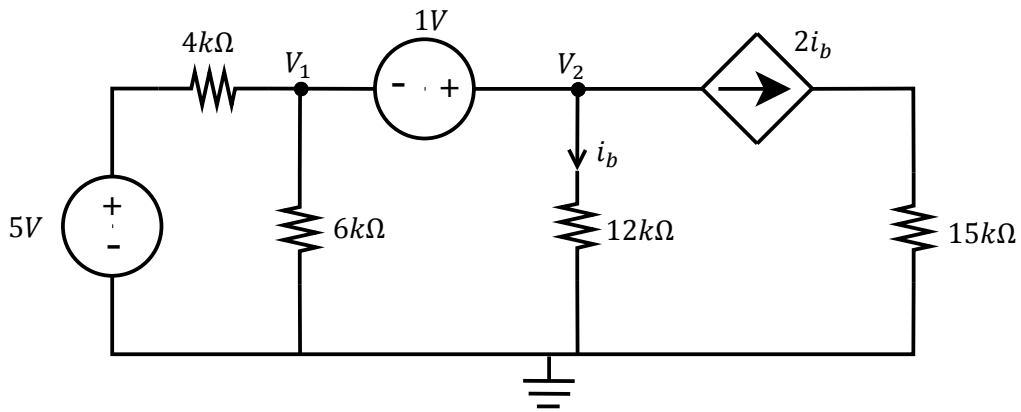
Gebruik node-analise om twee vergelykings van die volgende vorm op te stel:

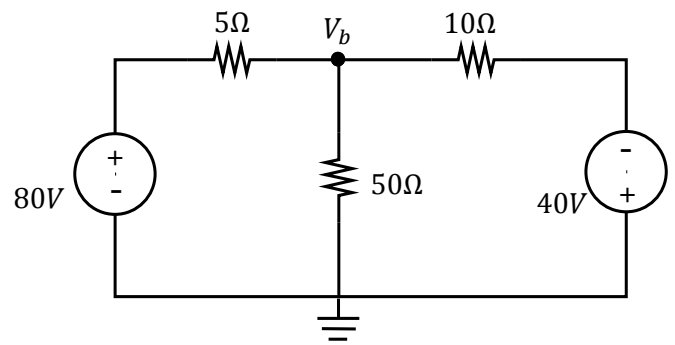
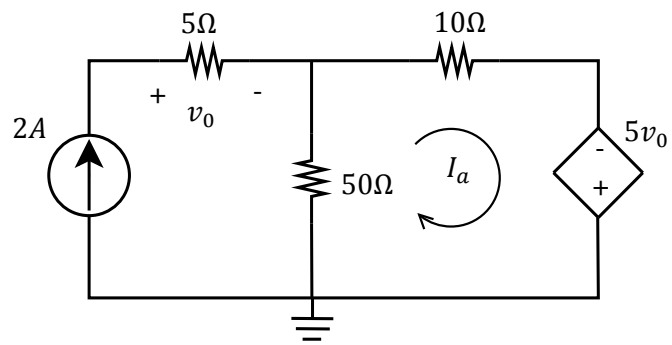
$$\begin{aligned} aV_1 + bV_2 &= -1 \\ 5V_1 + cV_2 &= d \quad (\text{KCL}) \end{aligned}$$

Question 2.1 [4]

Use nodal analysis to set up two equations of the following form:

$$\begin{aligned} aV_1 + bV_2 &= -1 \\ 5V_1 + cV_2 &= d \quad (\text{KCL}) \end{aligned}$$



Vraag 2.2 [4]Bereken I_a en V_b .**Question 2.2 [4]**Determine I_a and V_b .

Vraag 2.3 [4]

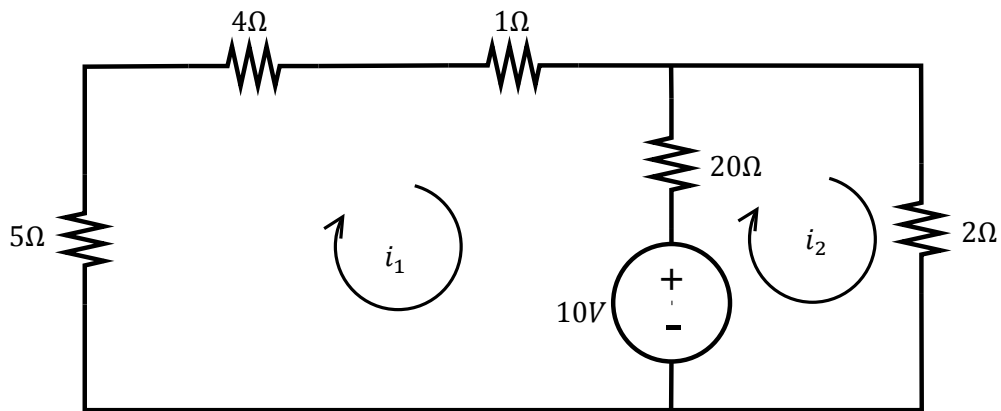
Gebruik lus-analise om twee vergelykings van die volgende vorm op te stel:

$$30i_1 + ei_2 = f$$
$$gi_1 + hi_2 = 10$$

Question 2.3 [4]

Use mesh analysis to set up two equations of the following form:

$$30i_1 + ei_2 = f$$
$$gi_1 + hi_2 = 10$$

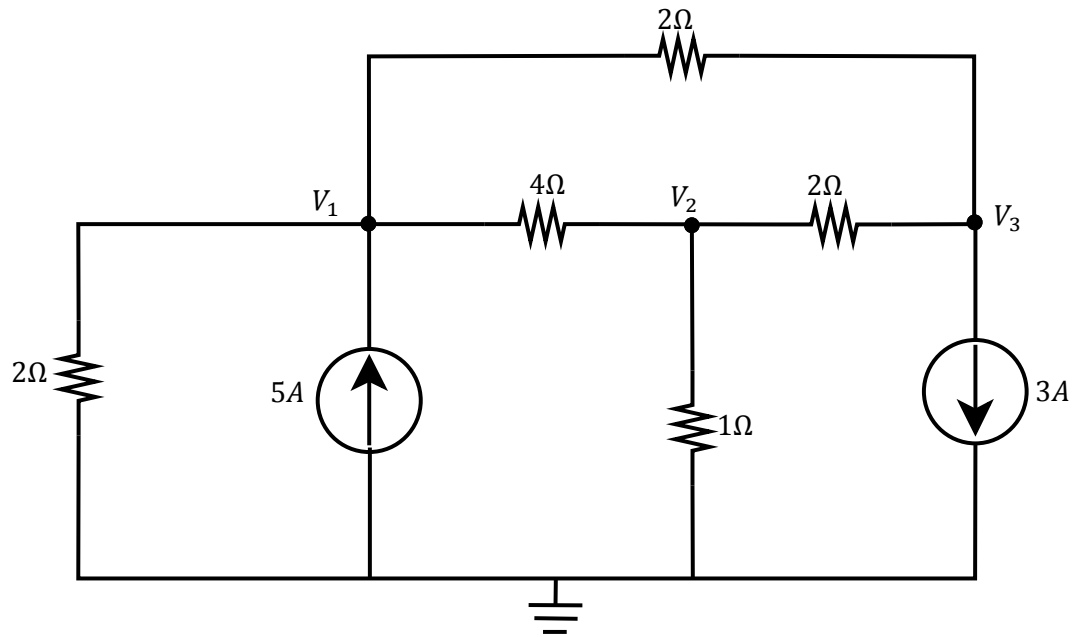


Vraag 2.4 [3]

Stel 'n matriksvergelyking d.m.v. inspeksie op in terme van die nodespannings.

Question 2.4 [3]

Use inspection to set up a matrix equation in terms of the node voltages.



Vraag 2.5 [4]a) Bepaal i_2 :

$$5i_1 - 10i_2 = 20$$

$$i_1 + i_2 = 2$$

b) Bepaal V_2 :

$$\begin{bmatrix} 5 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix}$$

Question 2.5 [4]a) Determine i_2 :

$$5i_1 - 10i_2 = 20$$

$$i_1 + i_2 = 2$$

b) Determine V_2 :

$$\begin{bmatrix} 5 & 0 & -2 \\ 0 & 2 & 0 \\ -2 & 0 & 1 \end{bmatrix} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 0 \\ -2 \\ 0 \end{bmatrix}$$

Vraag 2.6 [4]

Gegee: $\beta = 100$; $V_{BE} = 0.7$ V.

- a) Indien $V_S = 10$ V, bepaal I_B (in μA).
b) Indien $I_B = 5$ μA , bepaal v_o .

Question 2.6 [4]

Given: $\beta = 100$; $V_{BE} = 0.7$ V.

- a) If $V_S = 10$ V, determine I_B (in μA).
b) If $I_B = 5$ μA , determine v_o .

